EXPLOITS UNIVERSITY

BSc INFORMATION COMMUNICATION AND TECHNOLOGY

BICT 3202 · OBJECT-ORIENTED ANALYSIS AND DESIGN

WEEK 4

Assignment: The Building Blocks of UML

PROGRAMME	BSc Information Communication and Technology
COURSE CODE / YEAR	BICT 3202 · Year 3
LECTURER	Francis Fweta — ICT Lecturer
DELIVERY	2h Lecture · 1h Tutorial · 1h Practical · 10h Independent Learning

Prepared by Francis Fweta · Exploits University · Week 4 of 16

Contents

ASSIGNMENT INSTRUCTIONS

- Answer ALL questions in Sections A, B and C.
- Read the Week 4 module and your lecture notes before attempting each question.
- Draw diagrams neatly with correct UML notation.
- Cite OMG UML 2.5.1 where relevant.
- Submit by the date given by your lecturer. Total: 50 marks.

Section A — Short-Answer Questions (20 marks)

Question A1

Define a structural thing in UML. [2 marks]

Question A2

Define a behavioural thing in UML. [2 marks]

Question A3

Define a grouping thing in UML. [2 marks]

Question A4

Define an annotation thing in UML. [2 marks]

Question A5

What is an association relationship? [2 marks]

Question A6

What is a generalisation relationship? [2 marks]

Question A7

Differentiate aggregation from composition. [3 marks]

Question A8

What is a dependency relationship? [2 marks]

Question A9

Define multiplicity and give three examples. [2 marks]

Question A10

Explain how relationships help manage complexity in object models. [1 marks]

Section B — Application Questions (20 marks)

Question B1 (12 marks)

A university library system has Books, Members, Librarians and Loans.

- (a) Draw a class diagram showing these four classes with attributes. [4 marks]
- (b) Add associations between the classes with multiplicities. [4 marks]
- (c) Add one aggregation and one composition relationship with explanations. [4 marks]

Question B2 (8 marks)

Consider the relationship between Student and Course in a registration system.

- (a) What type of relationship is this? [2 marks]
- (b) What multiplicity would you assign and why? [2 marks]
- (c) How would this relationship appear in a class diagram? [2 marks]
- (d) What attributes might be needed to implement this relationship? [2 marks]

Section C — Case Study (10 marks)

'A hospital management system needs to model Patients, Doctors, Appointments, Departments and Medical Records.'

As an object-oriented analyst:

- (a) Identify all structural things. [2 marks]
- (b) Identify the relationships between these elements. [3 marks]
- (c) Draw a class diagram showing the complete structure. [3 marks]
- (d) Explain the business meaning of each relationship. [2 marks]

Marking Scheme

Section	Content	Marks
A	Short-answer questions (10 questions)	20
B	Application questions (12 + 8)	20
C	Case study	10
	TOTAL	50

Submission & Academic Integrity

- Submit a typed document (or neatly handwritten, as directed by your lecturer).
- This assignment is individual work. Plagiarism or copying will attract a penalty under university regulations.

- Cite any sources you consult using the format used in the Week 4 module references.